Lab: Secure AWS Account Setup + Free Tier EC2 Deployment

Objective: Set up a production-ready secure AWS account, understand infrastructure choices, apply Shared Responsibility, stay within Free Tier economics, while demonstrating core cloud computing principles.

Skills Covered:

- IAM & Account Security
- Billing & Cost Control
- Global Infrastructure (Region + AZ awareness)
- Shared Responsibility Model
- Cloud Economics (Free Tier + alerts)
- Console navigation & basic resource deployment

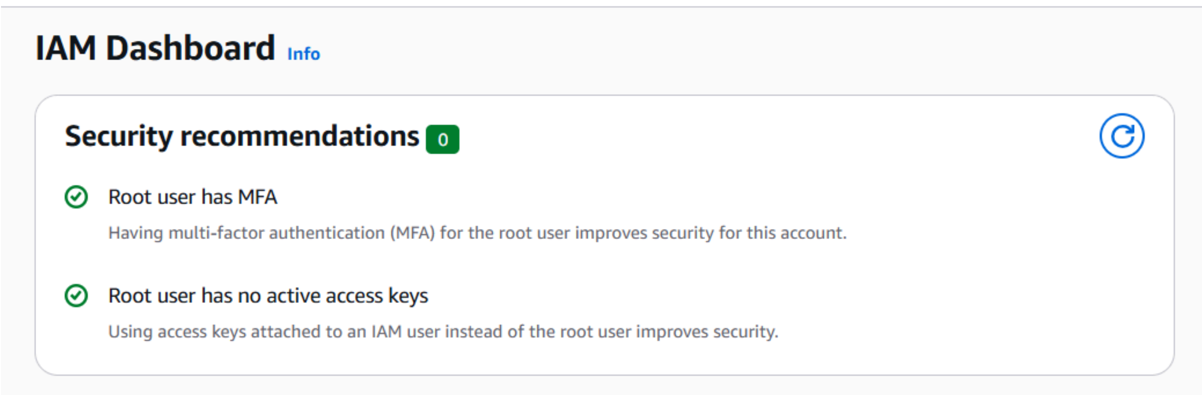
Pre-Lab Checklist

1. Log in as **Root User** (email + password)
2. Switch to **us-east-1** (N. Virginia) — critical for billing

Phase 1: Account Security & Billing Setup

Step 1: Enable MFA on Root Account (Non-negotiable)

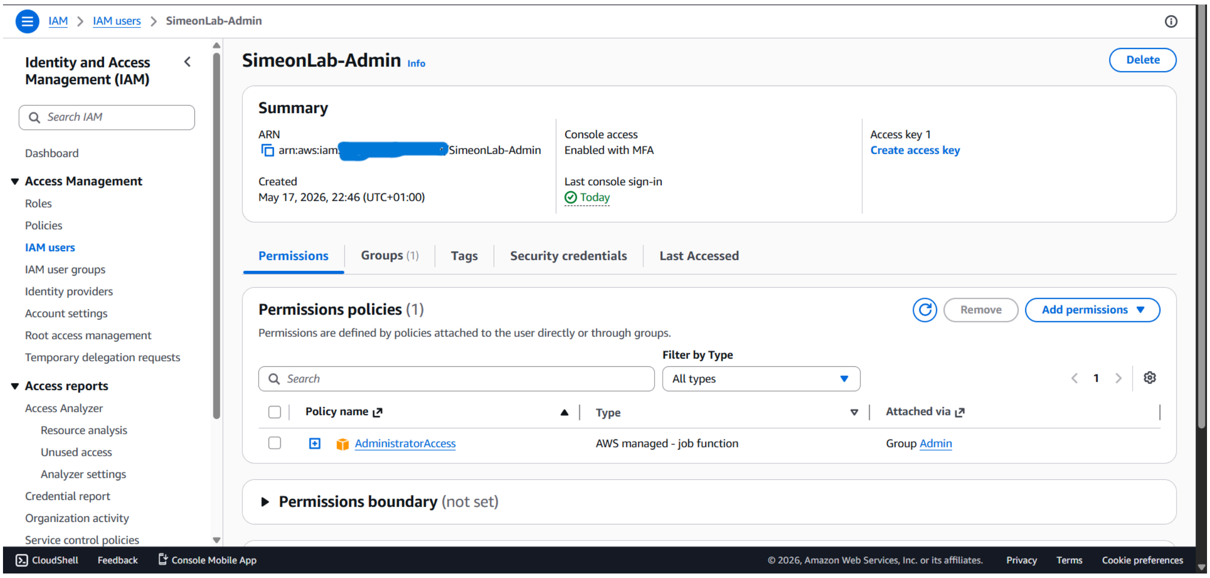
- Go to **IAM** → Your account name (top right) → **Security credentials**
- Under **Multi-factor authentication (MFA)** → Add MFA
- Use a virtual MFA app (Google Authenticator / Authy)

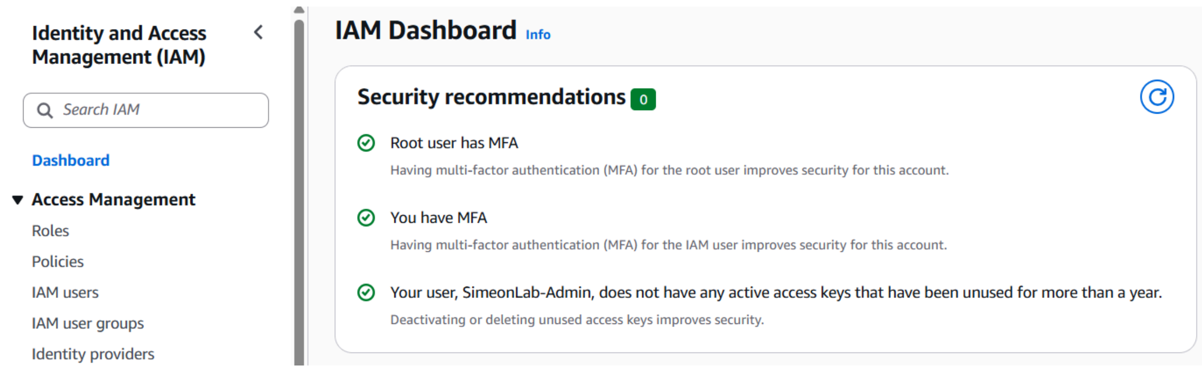


Step 2: Create an IAM Admin User

- IAM → Users → **Create user**
  - User name: YourName-Admin
  - Check **Provide console access**

    ○ Console password: Custom (temporary)
- Next → Attach policies: **AdministratorAccess** (for learning)

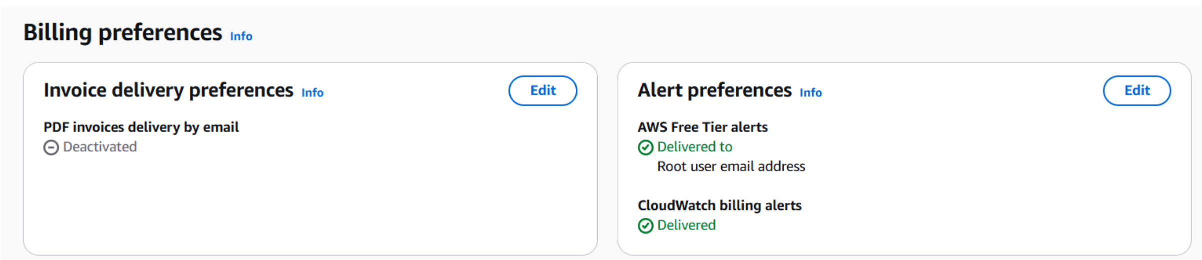




Step 3: Set Up Cost Protection

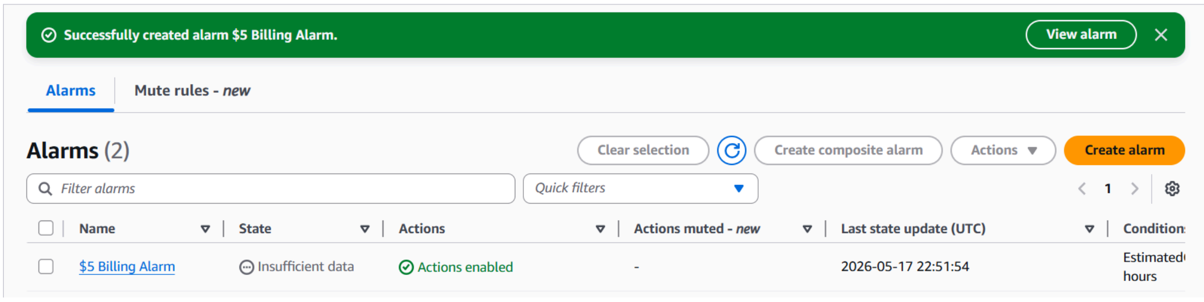
A. Free Tier Usage Alert

- Go to **Billing** → **Preferences** → Check **Receive Free Tier Usage Alerts**



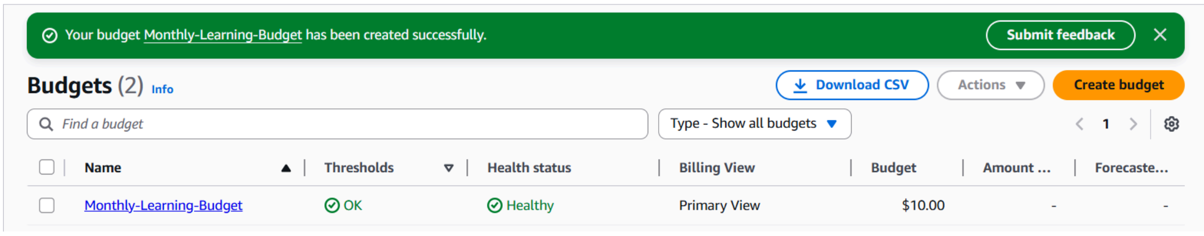
B. CloudWatch Billing Alarm ($5 threshold)

- Switch to **us-east-1**
- Go to **CloudWatch → Alarms → Billing → Create alarm**
  - Metric: EstimatedCharges
  - Statistic: Maximum
  - Threshold: Static → Greater than → 5
  - Create new SNS topic → Add your email → Create alarm
- Confirm subscription email from AWS



C. AWS Budgets (Recommended)

- Go to **Budgets → Create budget**
  - Type: Cost budget
  - Name: Monthly-Learning-Budget
  - Budgeted amount: 10 USD
  - Alerts: 50%, 80%, 100%



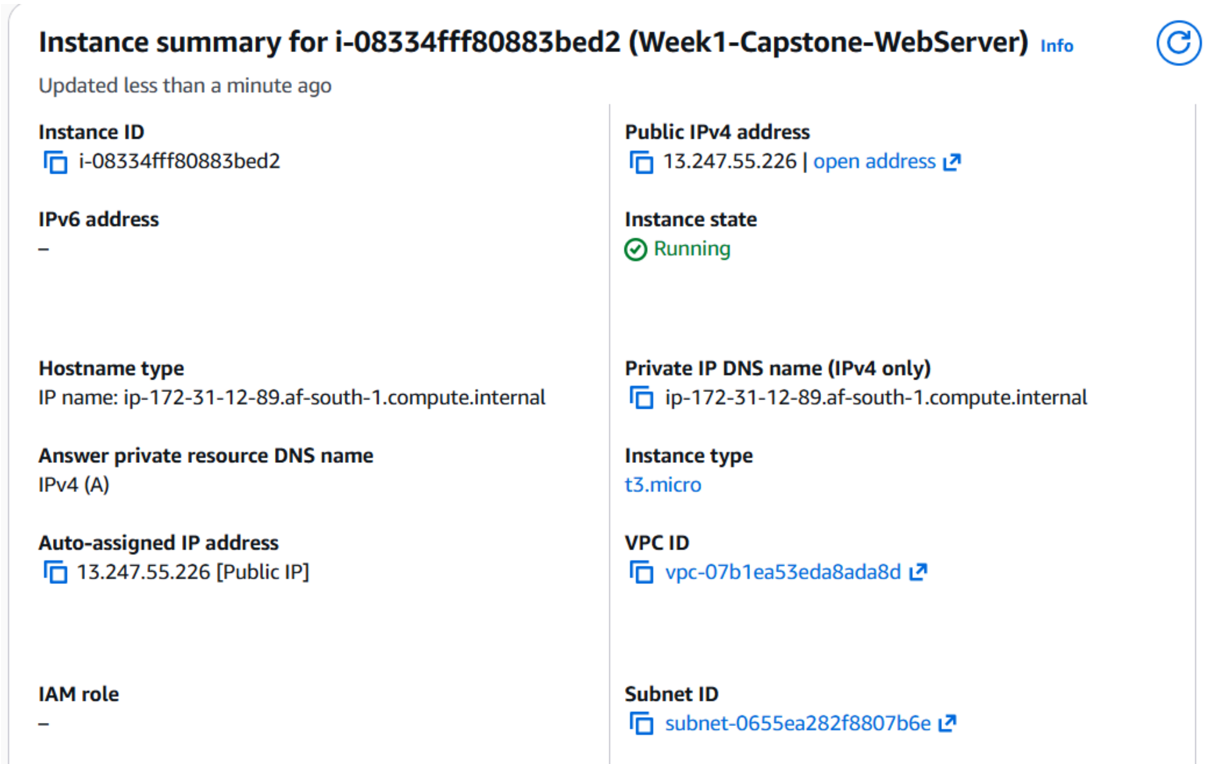
Verification: You should now have:

- MFA on Root
- IAM Admin user
- At least one billing alarm

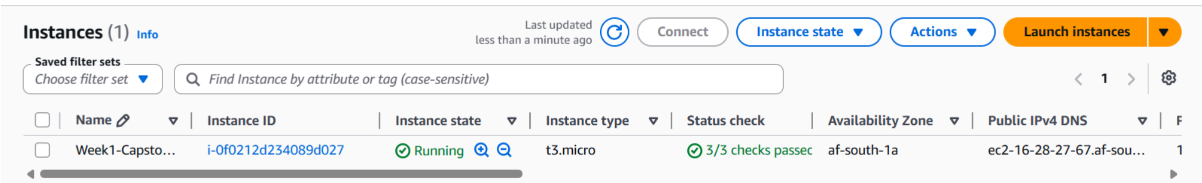
Phase 2: Infrastructure Awareness

Step 4: Launch a Free Tier EC2 Instance

1. Switch to **af-south-1** (Cape Town) — closest to Nigeria
2. Go to **EC2** → **Launch instance**
   - Name: Week1-Capstone-WebServer
   - AMI: **Amazon Linux 2023**
   - Instance type: **t3.micro**
   - Key pair: Create new (capstone-key)
   - Network settings:
     - Create new Security Group
     - Allow SSH (port 22) from **My IP** (for security)
     - Allow HTTP (port 80) from Anywhere (0.0.0.0/0) — for web server
   - Storage: Default 8GB gp3
   - **Launch**



3. Wait until **Status check = 2/2 checks passed**

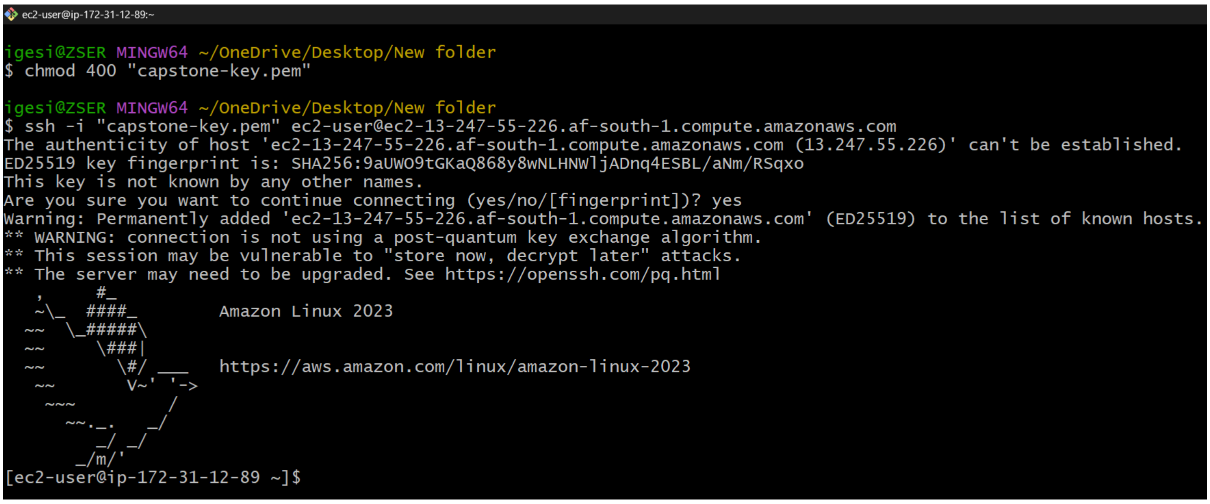


Phase 3: Connect & Configure (Hands-on Cloud)

Step 5: Connect to your instance

- Download the .pem key
- On your local terminal (Git Bash / WSL / Mac Terminal):

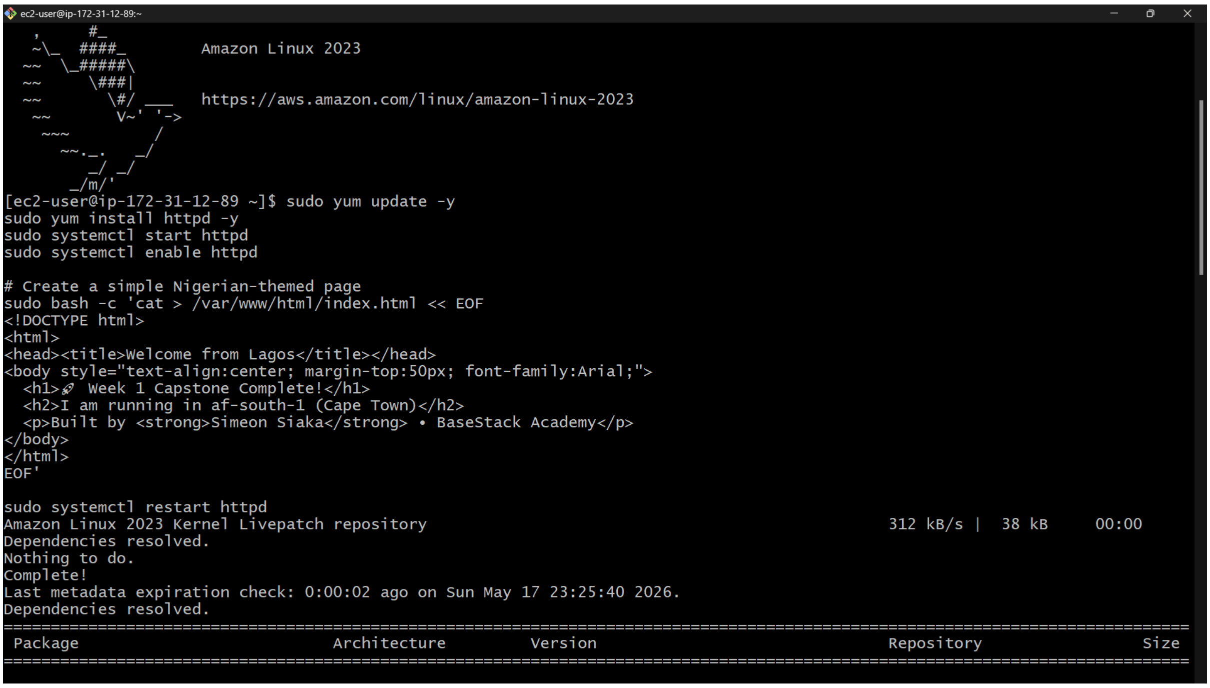
```bash
chmod 400 capstone-key.pem
ssh -i "capstone-key.pem" ec2-user@<your-public-ip>
```



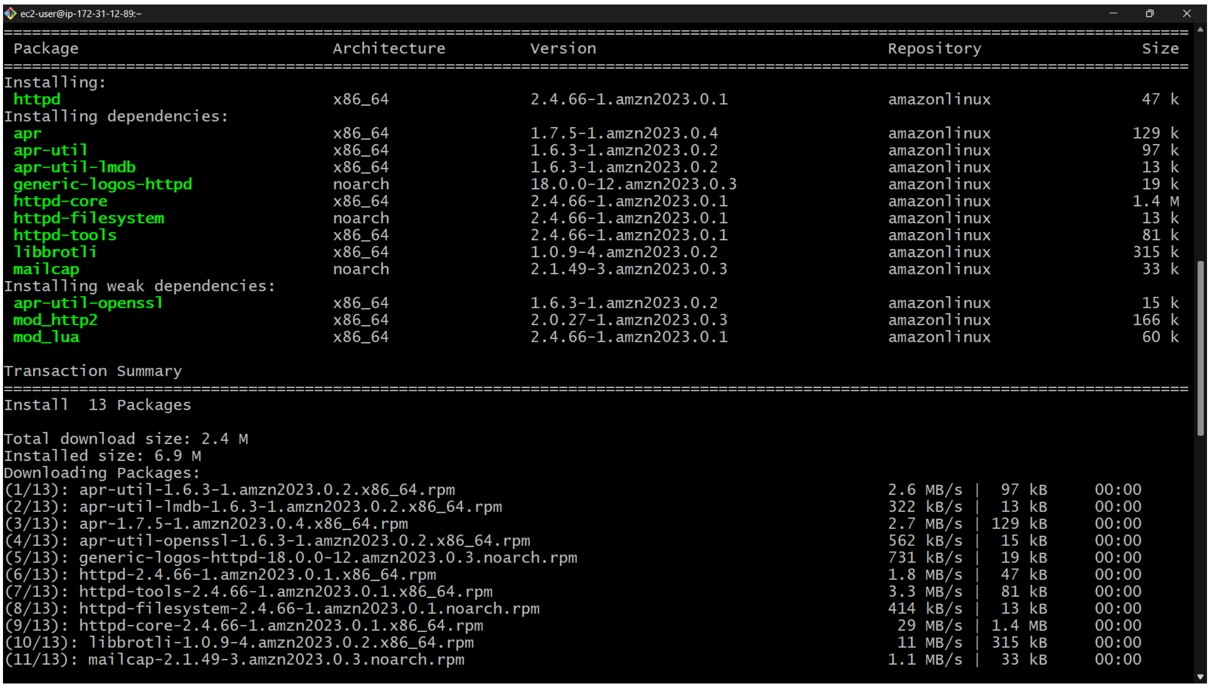
Step 6: Deploy a simple web server (Apache)

```bash
sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd

# Create a simple Nigerian-themed page
sudo bash -c 'cat > /var/www/html/index.html << EOF
<!DOCTYPE html>
<html>
<head><title>Welcome from Lagos</title></head>
<body style="text-align:center; margin-top:50px; font-family:Arial;">
  <h1>🚀 Week 1 Capstone Complete!</h1>
```

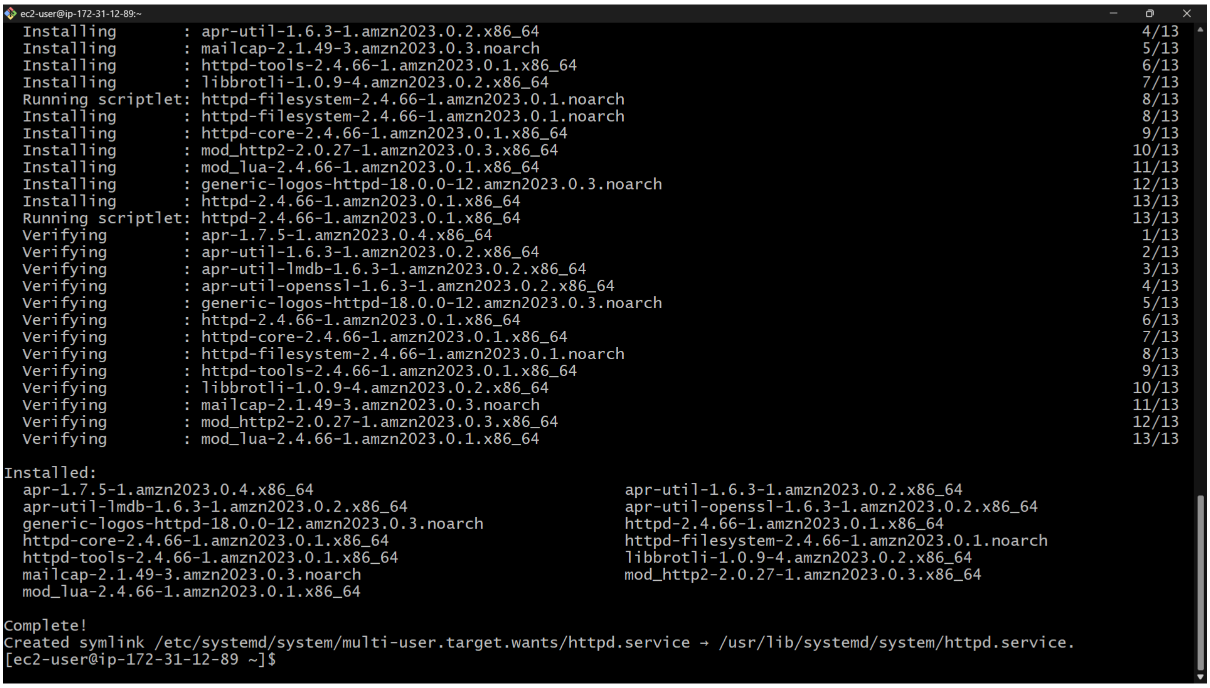
```
  <h2>I am running in af-south-1 (Cape Town)</h2>
  <p>Built by <strong>Your Name</strong> • BaseStack Academy</p>
</body>
</html>
EOF'

sudo systemctl restart httpd
```



```
ec2-user@ip-172-31-12-89:~
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-12-89 ~]$ sudo yum update -y
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd

# Create a simple Nigerian-themed page
sudo bash -c 'cat > /var/www/html/index.html << EOF
<!DOCTYPE html>
<html>
<head><title>Welcome from Lagos</title></head>
<body style="text-align:center; margin-top:50px; font-family:Arial;">
  <h1>🚀 Week 1 Capstone Complete!</h1>
  <h2>I am running in af-south-1 (Cape Town)</h2>
  <p>Built by <strong>Simeon Siaka</strong> • BaseStack Academy</p>
</body>
</html>
EOF'

sudo systemctl restart httpd
Amazon Linux 2023 Kernel Livepatch repository                312 kB/s |  38 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:02 ago on Sun May 17 23:25:40 2026.
Dependencies resolved.
================================================================================================
 Package                  Architecture     Version                    Repository          Size
================================================================================================
```



```
ec2-user@ip-172-31-12-89:~
================================================================================================
 Package                  Architecture     Version                    Repository          Size
================================================================================================
Installing:
 httpd                    x86_64           2.4.66-1.amzn2023.0.1      amazonlinux         47 k
Installing dependencies:
 apr                      x86_64           1.7.5-1.amzn2023.0.4       amazonlinux        129 k
 apr-util                 x86_64           1.6.3-1.amzn2023.0.2       amazonlinux         97 k
 apr-util-lmdb            x86_64           1.6.3-1.amzn2023.0.2       amazonlinux         13 k
 generic-logos-httpd      noarch           18.0.0-12.amzn2023.0.3     amazonlinux         19 k
 httpd-core               x86_64           2.4.66-1.amzn2023.0.1      amazonlinux        1.4 M
 httpd-filesystem         noarch           2.4.66-1.amzn2023.0.1      amazonlinux         13 k
 httpd-tools              x86_64           2.4.66-1.amzn2023.0.1      amazonlinux         81 k
 libbrotli                x86_64           1.0.9-4.amzn2023.0.2       amazonlinux        315 k
 mailcap                  noarch           2.1.49-3.amzn2023.0.3      amazonlinux         33 k
Installing weak dependencies:
 apr-util-openssl         x86_64           1.6.3-1.amzn2023.0.2       amazonlinux         15 k
 mod_http2                x86_64           2.0.27-1.amzn2023.0.3      amazonlinux        166 k
 mod_lua                  x86_64           2.4.66-1.amzn2023.0.1      amazonlinux         60 k

Transaction Summary
================================================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm               2.6 MB/s |  97 kB     00:00
(2/13): apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64.rpm          322 kB/s |  13 kB     00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                    2.7 MB/s | 129 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm       562 kB/s |  15 kB     00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm  731 kB/s |  19 kB     00:00
(6/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm                 1.8 MB/s |  47 kB     00:00
(7/13): httpd-tools-2.4.66-1.amzn2023.0.1.x86_64.rpm           3.3 MB/s |  81 kB     00:00
(8/13): httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch.rpm      414 kB/s |  13 kB     00:00
(9/13): httpd-core-2.4.66-1.amzn2023.0.1.x86_64.rpm             29 MB/s | 1.4 MB     00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm              11 MB/s | 315 kB     00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm              1.1 MB/s |  33 kB     00:00
```



```
  Installing       : apr-util-1.6.3-1.amzn2023.0.2.x86_64                                                    4/13
  Installing       : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                    5/13
  Installing       : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                                                6/13
  Installing       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                   7/13
  Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                                           8/13
  Installing       : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                                           8/13
  Installing       : httpd-core-2.4.66-1.amzn2023.0.1.x86_64                                                 9/13
  Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                                 10/13
  Installing       : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                                                   11/13
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                      12/13
  Installing       : httpd-2.4.66-1.amzn2023.0.1.x86_64                                                     13/13
  Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64                                                     13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                                                         1/13
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                                                    2/13
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                                               3/13
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64                                            4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                       5/13
  Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64                                                      6/13
  Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64                                                 7/13
  Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                                           8/13
  Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                                                9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                  10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                   11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                                 12/13
  Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                                                   13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                                  apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                        apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64                          httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                         libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                             mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-12-89 ~]$
```

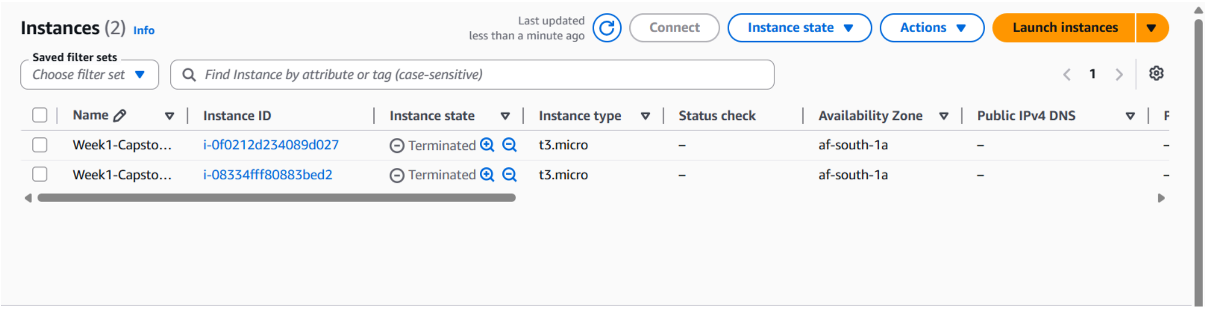
Open your browser: http://<your-public-ip>

You should see your page!

Phase 4: Cleanup + Economics Check

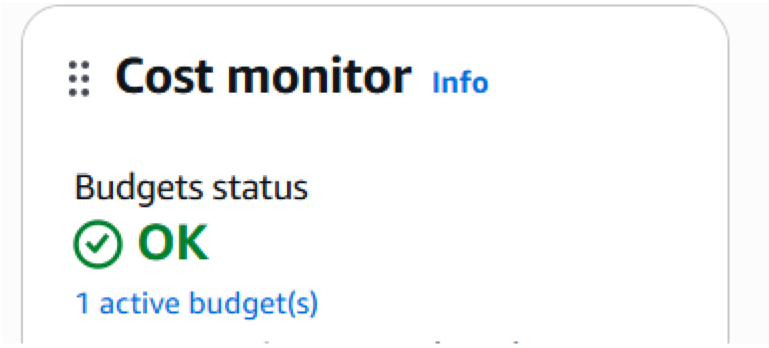
Step 7: Stop/Terminate Instance (Very Important)

- Go back to EC2 console
- Select instance → **Instance state** → **Stop** (you can start later)
- Or **Terminate** if you're done



Step 8: Final Cost Check

- Go to **Billing** → **Cost Explorer** (or **Budgets**)
- Confirm your spend is still near zero



Thank you